Proposition 0.1. *let* $\lambda = (\lambda_1, ..., \lambda_r)$ *and* $\mu = (\mu_1, ..., \mu_i)$

(1) $\exists k \in \mathbb{N}$ *such that* $\overline{\text{soc}}^n I^{\lambda,\mu} = 0$ *for all* $n > k$.

(2) also $k_{max} := \max_{\gamma_1,\gamma_2} |\gamma_1| + |\gamma_2|$ *is given by:* $k_{max} = \min\{|\lambda|, |\mu|\}$

Proof. Assume without loss of generality that $|\mu| < |\lambda|$, For sake of contradiction when proving (1), assume that there exists some $n \in \mathbb{N}$ such that $n > k_{max}$ with $\overline{\text{soc}}^n I^{\lambda,\mu} \neq 0$
this implies that

$$\bigoplus_{\lambda',\mu'} \bigoplus_{|\lambda|+|\mu|=n} N^\lambda \cdot N^\mu \neq 0$$

$$\iff \exists \gamma_1, \gamma_2 : |\lambda| + |\mu| = n \ \ s.t. \quad N^\lambda_{\gamma_1,\gamma_2,\lambda'} \cdot N^\lambda_{\gamma_1,\gamma_2,\mu'} \neq= 0$$

$$\iff N^\lambda_{\gamma_1,\gamma_2,\lambda'} \neq 0 \text{ and } N^\lambda_{\gamma_1,\gamma_2,\mu'} \neq 0$$

Well:

$$\max_{\gamma_1,\gamma_2} |\gamma_1| + |\gamma_2| := k_{max} < n = |\gamma_1| + |\gamma_2|$$

$$\iff \max_{\gamma_1,\gamma_2} |\gamma_1| + |\gamma_2| < |\gamma_1| + |\gamma_2|$$

which contradicts by deficinition of max.
For (2):
Look at any $k \neq 0$, that is $\exists \gamma_1, \gamma_2 : |\gamma_1| + |\gamma_2| = k \leq k_{max}$.

$$k = |\gamma_1| + |\gamma_2| \leq |\mu| = \min\{|\lambda|, |\mu|\}$$

$$\implies k \leq \min\{|\lambda|, |\mu|\}$$

Furtherermore given that $|\gamma_1| + |\gamma_2| = |\lambda| - |\lambda'|$ and $|\gamma_1| + |\gamma_2| = |\mu| - |\mu'|$. The fact that (1) guarantees existance of maximum means that this is precisely when $k = |\lambda| - |\lambda'| = |\mu| - |\mu'|$ is maximal. This is when $\mu' = \emptyset$ and $|\lambda| - |\lambda'| = |\mu|$.
$\implies |\gamma_1| + |\gamma_2| \leq k_{max} = |\mu| = \min\{|\lambda|, |\mu|\}$

$$\implies k_{max} = \min\{|\lambda|, |\mu|\}$$

□